Table 1: Finite-rank approximation errors and Ritz bounds

Application	Probe	p	R	Retained mass	p95 e_R	$\max e_R$	$\max B_R$
LaLonde ATE	overall ATE direction	11	5	0.999	0.007	0.011	0.805
Hurricane full fiscal/post probe	fiscal/post block probe	72	5	0.661	0.178	0.214	1.000 (vacuous)
Hurricane full fiscal/post probe	fiscal/post block probe	72	15	0.869	0.056	0.065	1.000 (vacuous)

Notes: e_R is the exact full-coordinate stress approximation error at $\lambda_0 = 1$ and B_R is the capped deterministic Ritz bound. The macro row uses the full-space temporal-resolvent benchmark built from 125-dimensional pre-PCA influence contributions; it is not a full-dimensional latent state-space estimate. Bounds are conditional on the estimated regularized covariance operators.

Table 2: Hurricane Ritz certification and severe-direction geometry

Panel A. Accuracy

Surface and rank	Probe	Trace share	Retained mass	p95 e_R	$\max e_R$	Share $e_R > .05$	p95 Ritz subspace residual term
72-coordinate full event-study surface, R=5	fiscal/post block probe	0.603	0.661	0.178	0.214	0.718	2.716
72-coordinate full event-study surface, R=10	fiscal/post block probe	0.723	0.759	0.135	0.144	0.574	4.298
72-coordinate full event-study surface, R=15	fiscal/post block probe	0.807	0.869	0.056	0.065	0.133	3.971
28-coordinate fiscal/post event-study surface, R=5	whole-surface soft probe	0.731	0.190	0.489	0.502	1.000	1.953
28-coordinate fiscal/post event-study surface, R=10	whole-surface soft probe	0.902	0.379	0.366	0.373	1.000	3.490
28-coordinate fiscal/post event-study surface, R=15	whole-surface soft probe	0.964	0.567	0.259	0.294	0.877	2.564

Panel B. Severe Geometry

Surface and threshold	Median d	Max d	Share $d > 0$	Median κ	p90 κ	Max generalized-eigenpair residual
72-coordinate full event-study surface, $a_\star = 2$	10.000	23	0.959	0.013	0.144	3.2e-13
72-coordinate full event-study surface, $a_\star = 4$	3.000	12	0.836	0.006	0.301	3.2e-13
28-coordinate fiscal/post event-study surface, $a_\star = 2$	2.000	15	0.621	0.108	0.337	1.4e-13
28-coordinate fiscal/post event-study surface, $a_\star = 4$	0.000	7	0.441	0.226	0.685	1.4e-13

Notes: The restricted surface is the true fiscal/post intersection derived from checked-in surface metadata. Panel A's residual term is $r_R(s)/\lambda$ with $\lambda = 1$, where $r_R(s)$ is the Ritz subspace residual. Panel B defines $\kappa_5(s; a_\star) = \|V_5' U_\star(s)\|_F^2 / d_\star(s; a_\star)$ when $d_\star > 0$. Interval eigenpairs are extracted with `scipy.linalg.eigh(..., subset_by_value=...)` and verified against full dense eigenvalue counts; the reported residual is the numerical generalized-eigenpair residual.

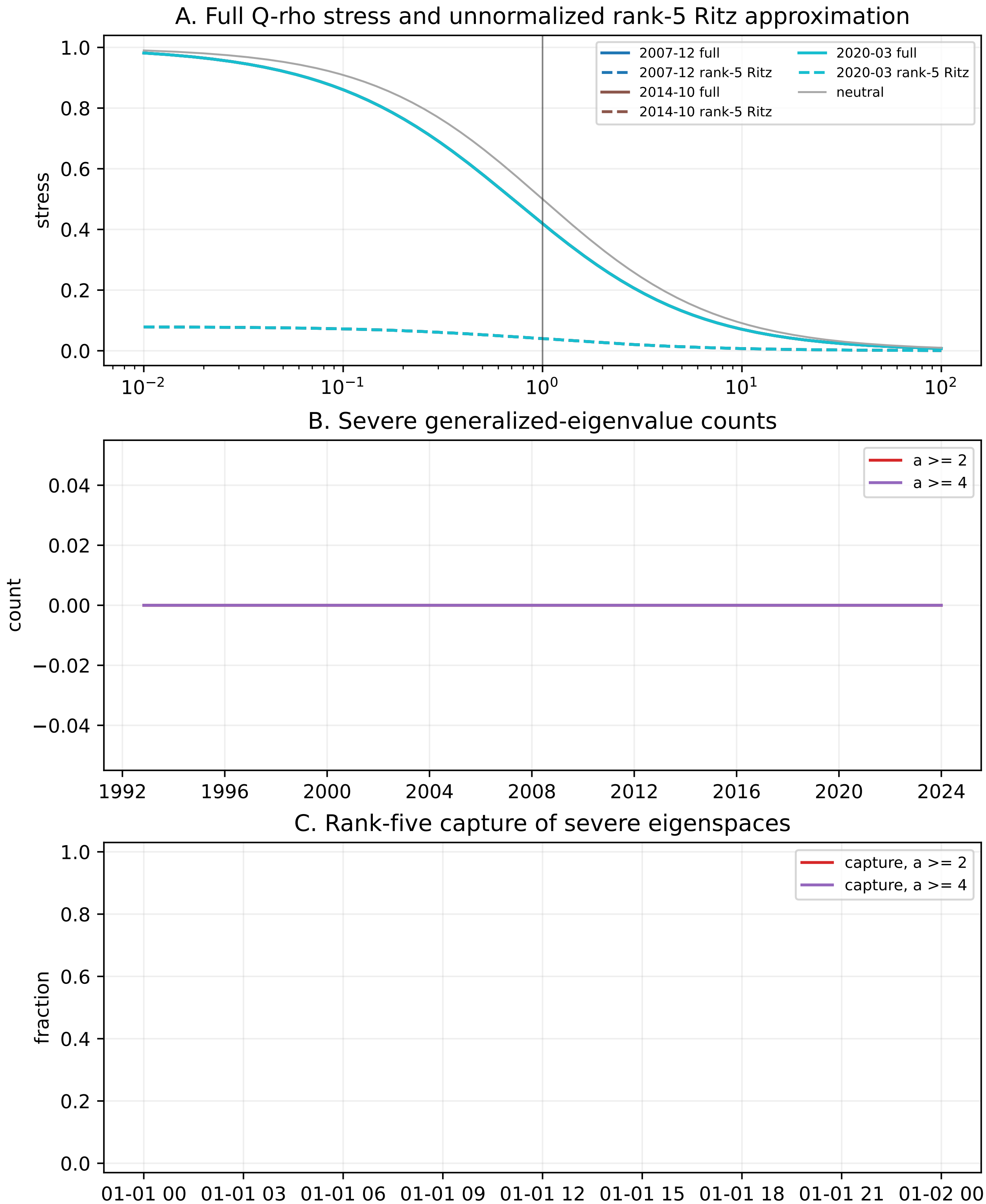

Figure 1: Macro Ritz interval-spectrum diagnostic.

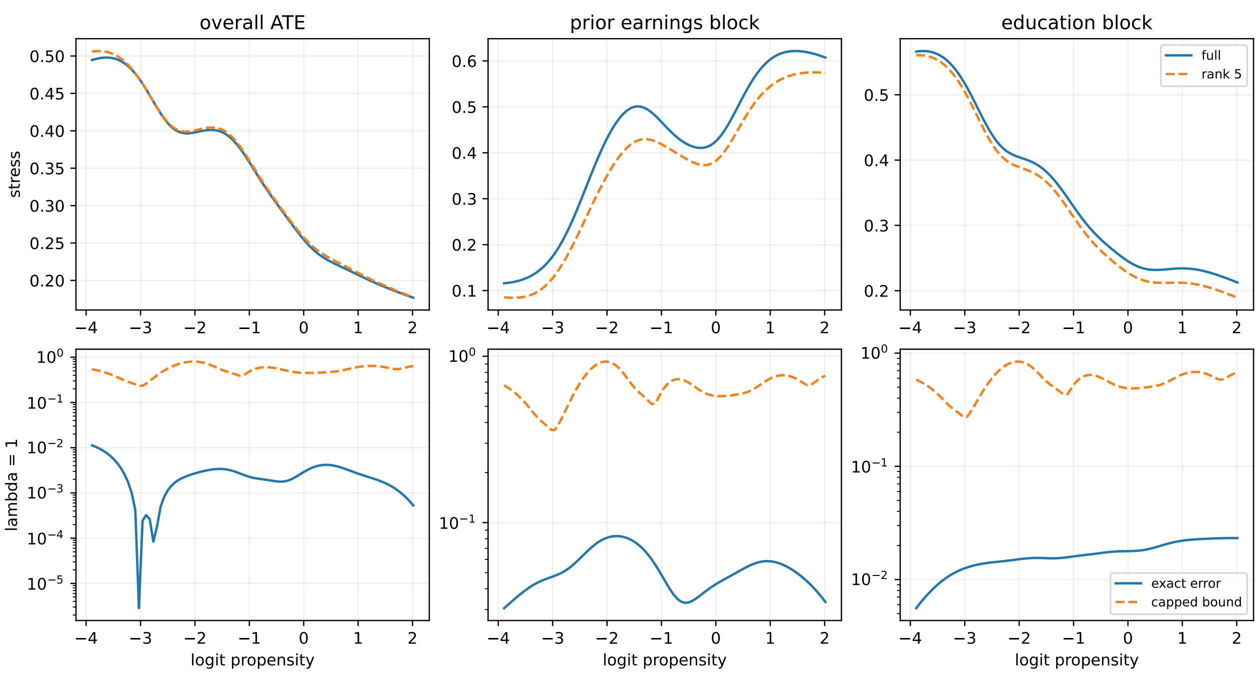

Figure 2: LaLonde Ritz validation.